

Problem 25.18

You are given

$$\int_0^n \bar{a}_{\bar{t}|} dt = 100.$$

Calculate $\bar{a}_{\bar{n}|}$.

Solution

For a continuous annuity,

$$\bar{a}_{\bar{t}|} = \frac{1 - v^t}{\delta}.$$

Therefore,

$$\int_0^n \bar{a}_{\bar{t}|} dt = \int_0^n \frac{1 - v^t}{\delta} dt.$$

This gives

$$\int_0^n \frac{1 - v^t}{\delta} dt = \int_0^n \frac{1}{\delta} dt - \int_0^n \frac{v^t}{\delta} dt.$$

Thus,

$$\int_0^n \bar{a}_{\bar{t}|} dt = \frac{n}{\delta} - \frac{1}{\delta} \int_0^n v^t dt.$$

Since $v = e^{-\delta}$,

$$v^t = e^{-\delta t}.$$

So

$$\int_0^n v^t dt = \int_0^n e^{-\delta t} dt = \frac{1 - v^n}{\delta}.$$

Therefore,

$$\int_0^n \bar{a}_{\bar{t}|} dt = \frac{n}{\delta} - \frac{1 - v^n}{\delta^2}.$$

Using

$$\bar{a}_{\bar{n}|} = \frac{1 - v^n}{\delta},$$

we get

$$\int_0^n \bar{a}_{\bar{t}|} dt = \frac{n - \bar{a}_{\bar{n}|}}{\delta}.$$

We are given that this integral equals 100, so

$$\frac{n - \bar{a}_{\bar{n}|}}{\delta} = 100.$$

Thus,

$$n - \bar{a}_{\bar{n}|} = 100\delta.$$

Therefore,

$$\boxed{\bar{a}_{\bar{n}|} = n - 100\delta}.$$